

One Way to Teach From Microeconomic Foundations I

At the Stanford Graduate School of Business, I typically have between 10 and 20 students enrolled in the course that is based on this book. We meet 20 times, for around 105 minutes per session. The students are first-year Ph.D. students from the GSB's economics and finance program, plus a small handful from other programs in the GSB, plus (in the past) 4 to 6 first-year Ph.D. students from the Economics Department who have already had a course on micro-theory out of Mas-Colell, Whinston, and Green (which is used as the text in the standard 1st year theory course in the Department) and who are (therefore) ready for something different. I will also pick up the occasional student from other places in the university; now that the book is out, I anticipate in particular some students from a program at Stanford in Financial Mathematics. (I gather my colleague Duffie is telling students in that program to take my course before taking his.)

The combination of small numbers and excellent preparation allows me to have the students do "presentations" in almost every class session. This worked extremely well last year (the first time I tried this), and I hope and plan to do so again this year.

I tell the students in advance that they may NOT use powerpoint or beamer or anything similar for these presentations; they should become comfortable using white board and markers, first. They are told to prepare seriously, in advance, an admonition that is supported by the fact that they are graded on their presentation. I suggest that serious preparation means that they should strive to present without reference to notes (although I do say that this is an ideal that is hard to achieve).

I do not give a final exam, per se. Instead, periodically (every two to three weeks) I assign, collect, and grade individually-done problem sets (where each "set" will consist of a small number of problem(s)). The students are graded based on how well these do this, on their oral presentations, and on general classroom participation. *Please note:* Concerning these specific problems, after writing and handing out a solution to the third problem set and, in particular, Problem 6, I decided to add to the *Student's Guide* the note now posted there on the Envelope Theorem. So assigning this problem no longer poses much of a challenge (if your students are using the SG at all).

Following are my tentative session-by-session assignments for this coming quarter. (They are tentative largely because I may need to make adjustments if I have more students than I anticipate.) Following that are the four do-by-yourself written assignments from last AY.

Session-by-Session Assignments

There are four types of activity that are assigned.

READ means read for full comprehension, which in turn means reading with a pad of paper and a pen/ pencil next to the book, where you write as you read. You might also want to have a highlighter handy. If there are things you don't understand, note them, and be ready to ask about them in class.

PREPARE means work through the problem, first alone, and then (as needed) collectively. Depending on what else we have to do in a given day, we may work through the problems assigned in this category in class; I am most likely to ask you if you had any difficulties with any of the assigned problems; if so, we'll concentrate on those.

SPEAKING ASSIGNMENT outlines the one-person-(or more, as indicated)-stands-to-and-delivers for each class. I would be grateful if, once you divvy up these assignments, you let me know who is assigned to which specific assignment.

DO-IT-YOURSELF ASSIGNMENTS consist of one or more problems that you do by yourself and turn in to me at the class session indicated. (You can email me a pdf if you prefer; be sure it arrives in my inbox before the indicated class session.) You may not discuss these assignments (until they are turned in) with anyone but me. But these are all open-book assignments—feel free to search the literature for answers, if you wish. If you are stuck on a problem, it is completely legit for you to come to me for a hint/nudge in the right direction.

The following list of assignments is tentative and subject to modification as we go along. Any changes will be announced via email.

Session #1 (Tuesday, September 24) :

READ the Preface to the Textbook and Chapter 1.

PREPARE Problems 1.5, 1.8, and 1.9, which should not prove difficult. Have a look at 1.11: If you are sure you can do it, fine; but if you aren't sure, it might be worthwhile to work your way through this. This sort of epsilon-and-delta real analysis is a huge part of what we'll be doing, and this problem is a good refresher if you haven't done this sort of thing for a while. Note that the solution of 1.11

is in the *Student's Guide*, so if you get stuck, you can look there for hints. If you have time and energy, you can also try 1.12 and 1.16—let me warn you that the solution to 1.16 in the *Guide* runs a full five pages, so don't expect to do this one in a paragraph or two. Finally, if you are feeling very good about your abilities at real analysis, give 1.13 a shot. This is actually fairly simple once you see what to do, so it is a cheat to look up the answer in the *Guide*.

SPEAKING ASSIGNMENT: Since you probably will not have time to organize yourselves, there is no official speaking assignment for session 1. Instead, I will demonstrate what I want by providing a mini-lecture on Problem 1.10, which is the standard counterexample to the (false) proposition that every complete and transitive binary relation admits a numerical representation. The assignment to which I will be responding is: Give a mini-lecture that (a) presents the problem in context, (b) outlines what needs to be shown in the solution, (c) presents whichever parts of the details of the solution you think are significant, and (d) (something not in the problem itself) explain why the lexicographic preferences given in this problem cannot be continuous, and then demonstrate by example that they are not.

Session #2 (Thursday, September 26) :

READ Chapter 2. Review Appendices 2 and 3 as needed.

PREPARE Problems 2.1, 2.4, 2.6, 2.10, and 2.13. If you find this to be relatively easy, try Problem 2.14, referring to the solution in the *Guide* as needed.

SPEAKING ASSIGNMENT: Present the solution to Problem 2.13. Your presentation should be complete, including definitions of things like “quasi-linearity.”

Session #3 (Tuesday, October 1) :

READ Appendices 4 and 5 and (then) Chapter 3.

PREPARE the two exercises given Appendix 4, then Problem 3.14, and then Problems 3.8, 3.9, and 3.11.

SPEAKING ASSIGNMENT: Present Problem 3.14. This means, give a mini-lecture on the problem: Set it up, and then provide answers to all parts. Be prepared to take questions on what goes into this problem—you don't need to have the proof of Berge's Theorem at your fingertips, but you probably should have a firm grasp on its statement. Also, note in part c of the problem that there is “more than one answer”; so a good presentation will not settle for just one way of doing this.

PLEASE NOTE: We are unlikely to cover all this in one day. I expect that we get as far as the end of Section 3.3 and the presentation of Problem 3.14, in which case the

remainder of this material will be done in Session 4.

Session #4 (Thursday, October 3) :

REVIEW as needed, the reading and problems assigned for Session 3 that we did not cover. We'll begin with this material.

READ in addition, Chapter 4.

PREPARE Problems 4.1 and 4.4.

SPEAKING ASSIGNMENT: Go back to Chapter 1 and review (or, read for the first time) Problem 1.16. As needed, review its solution provided in the *Guide*. In class, your speaking assignment is to compare and contrast Problem 1.16 (and, in particular, Proposition 1.23) with Afriat's Theorem. You should concentrate on the differences and similarities in the formulation of the two results; you should NOT regurgitate details of the proofs. But you get bonus points if you *connect* the proofs of the two (how are the logical flows different and / or the same); this is less important than explaining how the results themselves are the same and how they differ. DO NOT ignore part a of the Problem—an important point to be presented is what this is saying and how it connects to Afriat's Theorem.

Session #5 (Tuesday, October 8) :

READ Chapter 5. This is a long chapter (one of the two longest in the book), and we will definitely take two sessions on this material. So: begin by reading through the chapter without bothering about the details of proofs, to get a sense of the flow of the chapter. Then, for this class session, carefully read through the end of Section 5.2. The subsection "Expected Utility for Non-simple Lotteries" may be a challenge, but do the best you can with it. (Concerning this subsection, see the Speaking Assignment immediately following.)

PREPARE Problems 5.1 and 5.3.

SPEAKING ASSIGNMENT: If your book has a Problem 5.7 on page 122 (which, as I write this, is very unlikely), present its solution in class. If not, consult the *Student's Guide*, Chapter 5, specifically the discussion at the end concerning the proof of Proposition 5.10. The speaking assignment is to give a 30 minute (or so) lecturette on the proposition—why is it there, what is it saying, sketch of the proof, and so forth. Think of the assignment this way: Imagine you are giving this course, and you want to spend 30 minutes (only) on the issues connected to this Proposition. Your job is to choose what to say in the allotted 30 minutes (and what to leave out), to give the most (useful) information possible within the 30-minute window.

Please note: you shouldn't need to figure out anything technically: It is (mostly) all laid out for you in the *Student's Guide*. But 30 minutes isn't enough time to do everything, and the meat of this assignment is to use good judgment to decide on what you will say.

Session #6 (Thursday, October 10) :

READ the remainder of Chapter 5 carefully.

PREPARE Problems 5.4, 5.5, and 5.6.

SPEAKING ASSIGNMENT: Give a mini-lecture on Problems 5.4 and 5.5 and their connection to the Anscombe–Aumann developments.

DO-IT-YOURSELF ASSIGNMENT #1, which will be handed out in session #4, will be due today.

Session #7 (Tuesday, October 15) :

READ Chapter 6. This is a lot of material, and it isn't all on one point, so while I want you to read through the whole chapter, I want you to concentrate your attention on Section 6.1 as far as (but not including) the section on first- and second-order stochastic dominance, and then Section 6.2.

PREPARE Problems 6.2, 6.7, and 6.8. IF you are feeling under-challenged, I suggest you also try 6.5, 6.6, and 6.14.

SPEAKING ASSIGNMENT: Give a mini-lecture based on Problem 6.7. Extra credit: Suppose instead that the utility function in the problem was $U(x_1, x_2) = -e^{-x_1} - e^{-x_2}$, where we allow the consumer to consume negative amounts of either good. What can you say about the induced preferences for lotteries over y , where x_1 must be chosen before the uncertainty about y resolves?

Session #8 (Thursday, October 17) : This session and next Tuesday concern dynamic programming and dynamic choice. The emphasis on Thursday is dynamic programming, and the source material for this is Appendix 6, which is over 50 pages long, and no one would call it light reading. It would be great if you read and understood all the proofs, but your first order of business is to understand how the various propositions are used. So:

READ Appendix 6.

PREPARE: At the start of the chapter are seven problems. Most, but not quite all, of them are solved in the course of the chapter. Those not solved there are solved

in the *Guide*. What I want to do in class on Thursday is take the problems one at a time and discuss, How are they solved? Which propositions from the appendix are being used in the solution of each one, why, and how?

Speaking assignments for Session #8 and #9:

We will get to these after we do the PREPARE exercise for session 8, but before we tackle Chapter 7 itself. My guess is that we'll get to the first of these in session 8 and the second right at the start of session 9, but I want to be flexible on the timing. Hence, whoever is tasked with these two (one person for the first and another person for the second) should be ready to go in session 8. Each assignment has a time limit of 30 minutes. If you wish to use Excel on the computer to solve either/both numerically, you may do so, but you should NOT use powerpoint or beamer or similar in your presentation.

SPEAKING ASSIGNMENT 1: Present a solution to the following problem. At each date $t = 0, 1, 2, \dots$, a decision maker receives a payoff of either 2 or x_t , where $\{x_t\}$ is an i.i.d. sequence for which each x_t is either 2 or 1 or 0, each with probability $1/3$. The decision maker "chooses" whether to receive 2 or x_t , but she has a "budget" of times she can choose 2: She can choose 2 instead of x_t 33 times in the first 100 dates (that is, at dates 0 through 99), 33 times in the second 100 dates, and so forth. In every set of 100 dates, she has a budget of 33 times she can say, "I'll take 2 instead of x_t ." And she makes this declaration at date t (whether to use one of her budget or not) after seeing the value of x_t . Note: If $x_t = 2$, she does NOT have to use up one of her budget of times to declare for 2. She discounts her expected payoffs at rate 0.9 per period and seeks to maximize the expectation of the sum of her discounted payoffs. What is her optimal strategy?

SPEAKING ASSIGNMENT 2: Present a solution to the following problem. At each date $t = 0, 1, 2, \dots$, a decision maker receives a payoff of either 2 or x_t , where $\{x_t\}$ is an i.i.d. sequence for which each x_t is either 2, or 1, or 0, each with probability $1/3$. The decision maker's ability to declare that she wants 2 instead of x_t is limited as follows: She is either in state G or state P, and she can only declare for 2 if she is in state G. And, if she declares for 2, she goes into state P for the next 2 turns. That is, declaring for 2 when in state G causes the decision maker to lose the ability to declare for 2 for the next two turns. The decision maker seeks to maximize the expectation of the discounted sum of her rewards, discounted at a rate of 0.9 per period. What is her optimal strategy? (You must use the second half of Proposition A6.7.)

Session #9 (Tuesday, October 22): READ Chapter 7.

PREPARE Problem 7.2. Also read 7.3 and 7.4 and be sure you understand what the problems are asking for. You can try 7.3 if you have the energy and then 7.4 if you are still going (although I will not supply a written solution to 7.4, so if you want to see this proved, you will have to consult the relevant paper in the literature [which I wrote]).

Session #10 (Thursday, October 24) :

READ Chapter 8.

PREPARE Problems 8.1 and 8.11.

SPEAKING ASSIGNMENT: Present the solution to problem 8.11. It is imperative that you show how to use Excel to produce Figure 8.4 as part of your presentation. (You may use the in-class computer and projector to show how Excel does this, if you wish.)

Session #11 (Tuesday, October 29) :

READ through Chapter 9, to get a sense of the full “plot,” and then read more carefully through the end of Section 9.4, then section 9.8 and the first page and a bit of 9.9 (as far as the subsection “Cost Minimization for a Fixed Level of Output”).

PREPARE problems 9.1(a), 9.3, 9.4, 9.5, and 9.11. If you find all this easy, try 9.14.

SPEAKING ASSIGNMENT: Present the solution to Problem 9.3.

Session #12 (Thursday, October 31) :

READ the remainder of Chapter 9 carefully.

PREPARE Problems 9.1(b), 9.7, 9.8, and 9.13.

SPEAKING ASSIGNMENT: Present the solution to Problem 9.8.

DO-IT-YOURSELF ASSIGNMENT #2, which will be handed out in session #10, will be due today.

Session #13 (Tuesday, November 5) :

READ Chapter 10.

PREPARE Problems 10.2, 10.6, and 10.9. Warning: 10.9 will get a bit tricky at the end. It is a problem of the type “redo what happened in the chapter for the following variation.” If you have the plot line of the chapter down, you should be able to figure out the plot line for a full answer to this problem, even if you can’t do all

the detailed proofs. And getting the plot line down is more important. To reiterate a warning already given; everything follows the chapter nicely here until the very end, where it goes a little "weird" on you. So if you lose it at the end, it isn't a big deal.

SPEAKING ASSIGNMENT #1: The speaker for this session will lead off and present an overview of Chapter 10. What are the main results? How do they fit together? The proofs of propositions are not important, unless you feel that a proof contributes to the plot development. (A lot of the mathematics in this chapter and in Chapter 11 are more complicated versions of the math in Chapter 9. We will have gotten "down in the weeds" of the math in our discussion of Chapter 9; for Chapters 10 and 11, big-picture comprehension is more important. You have 30 minutes.

SPEAKING ASSIGNMENT #2: The one exception to what I just said is the proof of Proposition 10.9. Nothing like this appears in Chapter 9, and this is a proof technique that recurs in a few places, so I want to go over it carefully at least once. So, the second speaking assignment is, essentially, to present this proof, plus a bit more. Begin by stating the proposition. Then present the two counterexamples requested in problem 10.2. Then go through the proof of Proposition 10.9. 30 minutes at most, please.

Session #14 (Thursday, November 7) :

READ Chapter 11. As with Chapter 9, the plot line (which meanders quite a lot) is more important than the details. Indeed, when I come to the two "biggest" results in the subject, I punt on the details. If you have time to go back and absorb details of the proofs, fine. But don't worry if you don't have that time. (I am going to slight the use of the implicit function theorem, because there is too much else to cover.)

PREPARE: Start by constructing a one-side-of-a-sheet-of-paper outline of the plot of the chapter. Rather than having one of you present this as a speaking assignment, we'll construct this collectively to start class. Next do Problem 11.7. What are the terms in the Slutsky matrix for this alleged system of Marshallian demand? What is that telling you? Then try Problem 11.5. I'm more interested if you can figure out what is missing (which goes to your understanding of the plot line) than if you can fill in the details. So concentrate on that. And if you are chugging along nicely on all the details, try 11.3 and 11.4.

SPEAKING ASSIGNMENT: Present Problem 11.5. As the problem states, I'm less interested in how to "fix" the problem than in you recognizing what is the problem; what is missing? So in your presentation, concentrate on that: What do Propositions

11.4 and 11.9 say, and why don't they establish a clean duality, the way Proposition 10.17 does?

The final six sessions:

On Tuesday, November 12, we will cover Chapters 12 and 13. Then, beginning on Thursday and continuing for the final four sessions, we tackle General Equilibrium Theory, the last "unit" in the course and the book. Chapters 12 and 13 are already a lot of reading, but for the session on Thursday, November 14, I pile on the reading and work—I want you to read all of Chapter 14, supplementing with Appendix 8 (on fixed point theorems) as needed. And there is a fair bit of preparation work for the Thursday session, as well. And, if the pattern from years past is followed, students from the Department will have their 'metrics final somewhere in this rough window. Therefore, I strongly advise you to work ahead a bit over the weekend, doing at least some of the reading for Thursday's class and, perhaps, more.

Session #15 (Tuesday, November 12) :

READ Chapters 12 and 13. As with Chapters 10 and 11, your first and most important task is to understand the big picture of the two chapters. Once you have done that, you may begin to sweat the details.

PREPARE: Problems 13.2 and 13.6 are probably the most interesting, so have a look at those first, to the extent that you have the time and energy.

SPEAKING ASSIGNMENT: Present the solution to Problem 13.6. (Last year, the student doing this presentation asked the question: "Given that ζ^h is an excess demand function, why bother showing that it satisfies the weak axiom of revealed preference?" Whoever has this assignment might want to ponder this question and explain, as he or she presents, why the problem is set up the way it is.)

Session #16 (Thursday, November 14) :

READ Chapter 14, supplementing with Appendix 8 as needed. We will spread this material out over two sessions, with the existence stuff (probably) only being covered on Tuesday, so if you are pressed for time, read as far as the end of the Section 14.3 and then go on to the preparation of problems.

PREPARE Problems 14.2, 14.3, 14.4, and 14.5. On 14.5, it is enough to set it up; the numbers are very messy.

NO SPEAKING ASSIGNMENT, in view of all the other stuff you have to do. But be prepared with the four assigned problems: I'm going to expect you (all) to tell me how to proceed, as we work our way through these problems.

Session #17 (Tuesday, November 19) :

READ: Review as necessary Chapter 14 and Appendix 8. PREPARE Problems 14.7 through 14.11.

SPEAKING ASSIGNMENT 1: Give a mini-lecture on the solution to Problem 14.8.

SPEAKING ASSIGNMENT 2: Give a mini-lecture on the solution to Problem 14.9.

Both of these problems are solved for you in the *Guide*, so this is not a specially hard assignment at the level of finding the answer. What I will be looking for, though, in each case, is a polished presentation: give the set up, state the main result as a proposition, give the proof with details clearly explained. Imagine that you have to present this material to Ph.D. students who are seeing this sort of thing for the first time. You can assume that your audience understands all the definitions and Brouwer (for Problem 14.8) and Kakutani (for 14.9); you are showing them how to derive existence of Walrasian equilibrium in the somewhat easier context of an excess-demand function (instead of correspondence). You each have a time limit of 20 minutes.

Session #18 (Thursday, November 21) :

READ Chapter 15. If you only get as far as the end of Section 15.3, that's fine—we tackle the Debreu-Scarf theorem next class.

PREPARE Problems 15.2 and 15.5.

SPEAKING ASSIGNMENT: The proof of the First Theorem of Welfare Economics is a proof that you should be able to give on the spot, without notes, if awoken from a sound sleep at 3 am some morning. Get yourself to the point where you understand the proof that well. I will bring to class a “hat” or something similar with each of your names on slips of paper inside, and I will draw two names at random. The first name drawn will have to give a proof of the First Theorem on the board without looking at any notes. The second name will have to pick up the conversation at the end of the proof and explain why, in the proof, local nonsatiation of consumers is absolutely crucial to the result.

Over the Thanksgiving Break: I strongly suggest you take this week to read Chapter 16 carefully. It is particularly long, with a lot going on in it. If you leave it until after session #19, you will have a hard time with it.

Session #19 (Tuesday, December 3) :

READ (or reread) the remainder of Chapter 15 (beginning with Section 15.4). Then

read Chapter 16 through the end of Section 16.3.

PREPARE Problem 16.9 part a.

SPEAKING ASSIGNMENT: You will definitely need to coordinate on this one. There are at least 4 speaking parts (4 different people going to the board), and probably more.

Consider the following extended version of the Debreu-Scarf Theorem. In words:

Suppose we have a pure-exchange economy where the social endowment is strictly positive and each consumer has nondecreasing, strictly quasi-concave, and continuous utility. (Define what is meant by the N-replica version of this economy.)

(1) If a general allocation for the N-replica economy is a core allocation, it is an equal-treatment allocation.

(2) If a general allocation for the N-replica economy is a Walrasian equilibrium allocation, it is an equal treatment allocation.

(3) If we look at an allocation for the 1-replica economy and interpret it as an equal-treatment allocation in each N-replica economy, it is a Walrasian equilibrium allocation for some N-replica economy only if it is a Walrasian equilibrium allocation for all N-replicas.

(4) If an allocation for the 1-replica economy is interpreted as an equal-treatment allocation for each N-replica economy, and if it is a Walrasian equilibrium for the 1-replica economy, then it is a core allocation for every N-replica economy.

(5) If an allocation for the 1-replica economy, interpreted as an equal-treatment allocation for each N-replica economy, is in the core of each N-replica economy, then for some strictly positive price vector, it is a Walrasian equilibrium allocation for each N-replica economy.

That's the result. I want the class (collectively) to prove all this, with individuals taking turns at the board.

Speaker 1 will prove (1) and (2).

Speaker 2 will prove (3) and (4), then give the contrapositive form of (5) and connect (4) with (5)

Speaker 3 (who has the hardest job) will outline the "plot" for how (5) is provided. That is, this speaker, without going into details, will say "Step 1 in proving (5) is to show ...; step 2 is ...; ... How many steps are given, and how each step is

composed, is up to this speaker (although, in view of what comes next, he or she should do this in consultation with all of you). This speaker should not go into details of the proof, but should give information so that the audience understands what each step in the proof will be and how the steps knit together to prove (5).

Then speakers 4, 5, and so forth will give details of the proofs of each of the steps identified by Speaker 3. That is, speaker 4 will give details about step 1, speaker 5 will give details about step 2, and so forth, to the end.

At some point, someone must explain the diagram on the cover of the textbook. What is being depicted, there? It might be speaker 3, or one of the later speakers doing the various steps. Or you can designate a final speaker to cover this point.

Each speaker is responsible for answering questions on his/her part of the presentation. If this goes smoothly, you (collectively) get a special (secret) prize.

We'll begin the class session with this assignment. You should try to get it all done within 75 minutes.

DO-IT-YOURSELF ASSIGNMENT #3, which will be handed out in session #15, will be due today. (Because I expect you to be crunched with work just prior to Thanksgiving, I'm giving you the Thanksgiving holiday to work on this assignment.)

Session #20 (Thursday, December 5) :

READ (or reread) the remainder of Chapter 16.

PREPARE the rest of Problem 16.9 and then 16.12.

To celebrate the final day of the quarter, there is no speaking assignment. I believe the drill will be that you do course evaluations while in class this day; if so, you have to bring laptops. I'll give you more details about this when they are told to me.

(The four do-it-yourself written assignments from last AY are given beginning on page 14.)

Session #	Assignment	Speaker?
1	None--I demonstrate what I want	Kreps
2	Problem 2.13--homotheticity and quasi-linearity and indifference curve diagrams	
3	Problem 3.14--Robinson Crusoe solves a parametric optimization problem	
4	Afriat's theorem vs. similar issues in a more general context	
5	Discuss Proposition 5.10	
6	Problems 5.4 and 5.5 and the A-A trick	
7	Problem 6.7--late resolving uncertainty	
8 & 9	Assignment 1: A dynamic programming problem	
8 & 9	Assignment 2: Another d.p. problem	
10	Problem 8.11--Using the back half of Chapter 8 to solve a social welfare problem	
11	Problem 9.3--Berge's theorem in a neighborhood	
12	Problem 9.8--If one firm is more profitable than another (at all prices), is it uniformly more capable?	
13	Assignment 1: What is the plot line of Chapter 10?	
13	Assignment 2: The proof of Prop. 10.9, and two counterexamples related to the proposition	
14	Problem 11.5--What's missing in the chapter?	
15	Problem 13.6--The easy bits of Debreu's version of the S-M-D theorem	
16	None	
17	Assignment 1: Problem 14.8--Existence of W.e. with an aggregate excess demand function that goes all the way to the boundaries	
17	Assignment 2: Problem 14.9--and with an aggregate excess demand function with a boundary condition	
18	THESE TWO ASSIGNMENTS WILL BE COLD-CALLED. EVERYONE PREPARES Assignment 1: Prove (without notes) the 1st Theorem of Welfare Economics Assignment 2: Discuss the importance of local insatiability in the proof	Luck of the draw!
19	At least 4 speaking parts, and maybe more: See the assignment sheet for details	

Assignment #1

There are two problems. You should do both. I think problem 1 is pretty simple, but I could be wrong about this. I less confident about how hard is problems 2; but parts of it (at least) should be very easy.

■ 1. We discussed the lexicographic preference relation on $[0, 1]^2$, as defined in Problem 1.10 in class session 1, and then we discussed lexicographic preferences on R_+^k in class session 2. This problem concerns lexicographic preferences defined on R_+^2 , defined by $(x_1, x_2) \succeq_L (x'_1, x'_2)$ if (a) $x_1 > x'_1$ or (b) $x_1 = x'_1$ and $x_2 \geq x'_2$. We showed in class that $\succeq_L$ on the unit square is complete and transitive; you may *assume* that this is true as well for $\succeq_L$ on R_+^2 (which it is, using the same proof we gave in class). Recall that we also showed in class that there is no utility representation of these preferences.

(a) Let Q denote the set of rational numbers, Q_+ the set of nonnegative rational numbers, and Q_+^2 the set of nonnegative pairs of rational numbers. Consider $\succeq_L$ restricted to pairs (q_1, q_2) from Q_+^2 . Is there a utility function $u : Q_+^2 \rightarrow R$ that represents $\succeq_L$ on Q_+^2 ? Either prove that this is so, or prove that it is impossible.

(b) If you prove that it is possible, let u be a representing utility function, and extend u from Q_+^2 to R_+^2 by the rule

$$u((x_1, x_2)) = \sup \{u((q_1, q_2)) : (q_1, q_2) \in Q_+^2, (x_1, x_2) \succeq_L (q_1, q_2)\}.$$

We know that u , extended in this fashion, cannot give a representation of $\succeq_L$ on all of R_+^2 , because no such representation exists. But what, in particular, goes wrong? Be as clear and precise as you can concerning why u , extended in this fashion to R_+^2 , does not give a representation of $\succeq_L$ on all of R_+^2 .

(Note: If you conclude in part a that no representation on Q_+^2 is possible, you don't need to do part b. You might conclude that this is a strong hint that the answer to part a is, Yes, a representation is possible. But you still need to prove this.)

■ 2. Chapter 3 explores the intuition of bangs-for-the-buck in simple maximization problems with one linear constraint (plus nonnegativity constraints). And, in particular, it demonstrates (more or less) that the multiplier on the budget constraint is the “shadow value” of the consumer’s income. I want to get to the same result, but when we add another linear constraint.

The setting is the standard consumer problem, but with an additional constraint representing a maximum calorie count. Formally, there are k goods and the con-

sumption space is $X = R_+^k$. The consumer wishes to maximize her utility $u(x)$ subject to a budget constraint $p \cdot x \leq y$ for strictly positive price vector $p \in R_{++}^k$ and strictly positive income $y > 0$ and, in addition, subject to the constraint $c \cdot x \leq c^*$ for some $c \in R_{++}^k$ and $c^* > 0$. We also constrain x to be nonnegative.

You may assume throughout that u is strictly increasing, strictly concave, and twice continuously differentiable, and that c and p are not collinear. (I think it is enough that u is once continuously differentiable, but to be safe, may it 2x.)

(a) What are the first-order, complementary-slackness conditions for optimality? Comment (briefly) on whether these conditions are necessary and/or sufficient for optimality.

(b) The solution is unique. In one sentence, say why. (You don't need to prove this.)

(c) As we vary the parameters (p, c, y, c^*) , the solution varies continuously. In one sentence, say why. (You don't need to prove this.)

(d) Suppose I give you parameter values (p, c, y, c^*) , and I tell you that for an open set of parameter values around this vector of parameters, (i) the solution x^* is always strictly positive and (ii) satisfies both the budget and calorie constraints with equality. Suppose I increment y by an amount δ that is small enough to keep us inside this open set. To a first-order approximation, how should the consumer spend her extra income (to maximize utility)?

(e) Let λ be the multiplier on the budget constraint. I assert that, to a first-order approximation, if y is increased by δ , the consumer's (optimal) level of utility increases by $\lambda\delta$. Prove this. (Note: You do not need to solve part d to do part e, so even if you get stuck on d, try this part.)

Assignment #2

The problems are numbered 3 and 4, as part of a sequence that began with the first assignment.

■ 3. Recall that, in class, we discussed the following issue: We imagine that a consumer has preferences $\succeq$ over simple lotteries where the prize space $Z = R_+^k$; with the interpretation that a $z \in Z$ is a bundle of consumption goods. If we suppose that $\succeq$ satisfies the three mixture-space axioms, we know that $\succeq$ can be represented by expected utility, computed for some utility function $U : Z \rightarrow R$.

The issue is, What does it take in terms of assumptions on preferences to show that U is a continuous function on Z ? We showed (or, at least, I asserted) that a sufficient assumption is *If $\{z_n\}$ is a sequence of points in Z with limit z , and if $\delta_{z_n} \succeq p$ [respectively, if $p \succeq \delta_{z_n}$] for all n and some simple lottery p over Z , then $\delta_z \succeq p$ [respectively, $p \succeq \delta_z$].*

Can we weaken this condition? To show that U is a continuous function, is it sufficient to assume that *If $\{z_n\}$ is a sequence of points in Z with limit z , and if $\delta_{z_n} \succeq \delta_{z'}$ [respectively, if $\delta_{z'} \succeq \delta_{z_n}$] for all n and some $z' \in Z$, then $\delta_z \succeq \delta_{z'}$ [respectively, $\delta_{z'} \succeq \delta_z$].*

Either prove that this is sufficient, or give a counter-example if it is not.

■ 4. Consider the following optimal stopping problem. A random process $\{X_t; t = 0, 1, \dots\}$ begins with $X_0 = 1$ and, subsequently, either moves up 1 or down 1 at each date, moving up 1 with probability α and down 1 with probability $1 - \alpha$. (This moving up or down is independent of any earlier random events.) There is one exception to this rule: If ever $X_t = 0$, then $X_{t+1} = 0$ for sure, so 0 is “absorbing.”

A decision maker who is observing this process is allowed at any date (after seeing X_t) to say “pay me now,” in which case she gets X_t dollars, for whatever is the value of X_t . But she is only allowed to say this once. Her overall reward is the expected value of $(0.9)^T X_T$, where T is the time at which she says “pay me now”; if she is using a strategy in which there is positive probability that she never says “pay me now,” she gets 0 reward on that event. Note that her reward is discounted—she prefers to be paid earlier to later, if the reward is the same (in current value).

I may have made a mistake in my algebra, but if I did not, this is not an easy problem to solve in general. So here are two relatively easier exercises.

(a) For which values of α is the optimal strategy to stop the first time $X_t \geq 2$ (or when the process is absorbed in state 0)?

(b) For which values of α is the optimal strategy to stop the first time $X_t \geq 5$. (Note, I was unable to do this problem analytically; my answer will be based on numerical calculations.)

You must fully justify your claims.

Assignment #3

The problems are numbered 5, 6, and 7, as part of a sequence that began with the first assignment.

■ 5. In this problem, we want to clarify a few things about local insatiability and expenditure functions. To keep matters relatively simple, let me assume that u is a continuous and globally insatiable utility function, and e is the expenditure function generated by u . Also, so you can freely cite results from the text, assume that $u(0) = 0$ and $\sup\{u(x) : x \in R_+^k\} = \infty$.

(a) (Worth only one point): We know that if u is locally insatiable, $(p, v) \rightarrow e(p, v)$ is a continuous function. Why?

(b) Suppose that $(p, v) \rightarrow e(p, v)$ is continuous. Is u necessarily locally insatiable? Is $\hat{u}$, where $\hat{u}$ is generated from u in the fashion of Proposition 10.13?

(c) True or false? u is locally insatiable if and only if $\hat{u}$ is locally insatiable. I want you to tell me about both halves of this if and only if. Is it in both directions? True in one but not the other? False in both? A mere assertion is not enough—if you think it is true in either direction, you must give either a proof or cite a result from the text. If you think it is false, you must produce a counterexample and convincingly argue that your counterexample is what you say it is.

(d) How does indirect utility fit into this picture? If u is continuous and globally insatiable, $\hat{u}$ is defined from u via Proposition 10.13, and ν is the indirect utility function generated from u , what sort of conclusions can you draw about “local insatiability” (of either u or $\hat{u}$) and properties of ν ?

WARNING: This is more of a “find-it-in-the-book” question than a “invent-it-on-your-own” question. Nearly everything (but not quite everything) you need to know can be found somewhere in Chapters 10 and 11. So if this seems pretty easy to you, don’t go looking for mysteries.

■ 6. In the book (and in class), I avoid referring to something called “The Envelope Theorem,” because that name gets attached to a variety of specific results. But, as part of your education, you should know something about the general class of results associated with this name. In fact, you have already seen three specific examples of this general result. In this problem, I ask you to prove a generalization (sort-of) of two of those results. You are to prove this from first principles; you should not simply cite a result from some book. You can, if you wish, search the literature for proofs (although the precise result I’m asking you to prove is not the

“industry standard,” so you may have a problem finding a proof), but you have to give the details in your answer, and if you use a source other than the textbook, you must cite it. (You are allowed to assume standard results from analysis, such as—hint, hint—the mean-value theorem.) *You may not add assumptions to the ones I provide.* Here is what you must prove:

Consider the problem

Maximize $F(x, \theta)$ in x , for fixed θ , subject to $x \in X$.

Assume that $F : R^n \times R^m \rightarrow R$, where $X \subseteq R^n$ is compact. Let $f(\theta)$ be the maximized value; that is,

$$f(\theta) := \sup \{F(x, \theta); x \in X\}, \text{ for each } \theta \in R^m.$$

Assume that F is jointly continuous in x and θ and that F is continuously differentiable in θ ; assume further that the gradient of F with respect to θ is jointly continuous in x and θ . Prove that if, for a given θ^0 , there is a unique solution to the problem x^0 , then f is differentiable at θ^0 , and

$$\left. \frac{\partial f}{\partial \theta_i} \right|_{(\theta^0)} = \left. \frac{\partial F}{\partial \theta_i} \right|_{(x^0, \theta^0)}.$$

■ 7. In Chapter 9, when we studied the theory of the firm, we modeled the production-possibility set of the firm by a set Z and we assumed there is (otherwise) a perfectly competitive market in every good (input or output), with prices given by a strictly positive vector $p \in R_{++}^k$.

But suppose that some factors of production are not marketed at all. Model this as follows. We let k be the number of goods (inputs and outputs) for which there are perfectly competitive markets, and we let $p \in R_{++}^k$ be the vector of (strictly positive) prices for those goods, with $z \in R^k$ a netput vector for the firm.

But in place of Z , we imagine a function $F : R^k \rightarrow R^n$ for some integer n , we imagine that the firm is endowed with a vector $\alpha \in R^n$ of *resources*, and given the firm's resource vector α , the firm can choose any $z \in R^k$ that satisfies

$$F(z) \leq \alpha.$$

The firm, we will assume, places zero value on preserving its resources; it seeks to maximize its profit subject to its resource constraint, so in this setting the firm's

profit maximization problem is

$$\text{maximize } p \cdot z, \quad \text{subject to } F(z) \leq \alpha.$$

Assume that F is continuous, nondecreasing, and strictly increasing for strict increases in z . (Please note carefully: vectors z can have negative components, and F being increasing means higher values of F [more resources needed] for smaller [closer to zero] levels of input and for larger [further from zero] levels of output.)

There is no guarantee that, for an arbitrary vector $\alpha \in R^n$, some $z \in R^k$ satisfies $F(z) \leq \alpha$. So let $A = \{\alpha \in R^n : F(z) \leq \alpha \text{ for some } z \in R^k\}$. It is obvious that, if $\alpha \in A$ and $\alpha' \geq \alpha$, then $\alpha' \in A$. You may assume that A is nonempty.

- a. Prove that A is an open set and that an alternative characterization of A is: $A = \{\alpha \in R^n : F(z) < \alpha \text{ for some } z \in R^k\}$, where by $<$ we mean strictly less in every component.
- b. For $\alpha \in A$, let $Z(\alpha) := \{z \in R^k : F(z) \leq \alpha\}$. Prove: If F is a convex function, then $Z(\alpha)$ is a convex set for each $\alpha \in A$. Also prove: If F is a convex function, then A is a convex set in R^n .

(Appendix 3 only defines convexity of real-valued functions, so let me add here: $F : R^k \rightarrow R^n$ is **convex** if, for all z and $z' \in R^k$ and $a \in [0, 1]$, $F(az + (1 - a)z') \leq aF(z) + (1 - a)F(z')$, where $\leq$ means less or equal in the vector sense.)

- c. Fix $\alpha \in A$. What are necessary and sufficient conditions for: The problem $\max p \cdot z$, subject to $F(z) \leq \alpha$, has a finite solution for every $p \in R_{++}^k$.

Henceforth, assume whatever condition you gave in part c. (Or, if you were unable to do part c, simply assume: For every $\alpha \in A$ and $p \in R_{++}^k$, the problem $\max p \cdot z$, subject to $F(z) \leq \alpha$, has a finite solution.)

Let $Z^*(p, \alpha)$ be the set of solutions to this problem, fixing p and α , and let

$$\pi(p, \alpha) = \max \{p \cdot z : F(z) \leq \alpha\}.$$

(Note that it is legitimate to have a *max* here, since we are making assumptions sufficient to guarantee existence of a solution.)

- d. Prove that π is convex in p for each fixed $\alpha \in A$.
- e. Suppose that $z \in Z^*(p, \alpha)$. Is it true that $F(z) = \alpha$? (Do all the constraints necessarily bind?) Either prove or give a counterexample.

- f. I would like to “Berge” this problem. This means, I would like to show that $(p, \alpha) \rightarrow \pi(p, \alpha)$ is continuous and $(p, \alpha) \Rightarrow Z^*(p, \alpha)$ is upper semi-continuous and locally bounded. Tell me what you can about the “Berging” of this problem. Can it be done? Is it (perhaps) true on a restricted domain of prices and resource vectors α ? Do you need additional assumptions to make it true? I want you to give me as clear an exposition on this matter as you can.
- g. Give conditions under which π is concave in α for each fixed p . So this makes sense, you may assume that F is convex, so that A is a convex set. (If you couldn’t do part b, you may assume here that F is a convex function and that A is a convex set.)
- h. Using whatever conditions you came up with in part g to ensure that π is concave in α for fixed p , fix an $\alpha^0 \in A$ and let $(\beta, \gamma) \in R^n \times R$ be a supergradient of π in α at α^0 for the fixed p . That is,

$$\beta \cdot \alpha^0 + \gamma = \pi(p, \alpha^0) \quad \text{and} \quad \beta \alpha + \gamma \geq \pi(p, \alpha) \quad \text{for all } \alpha \in A.$$

Prove that $\beta \geq 0$ and, if z^* is a solution to the problem at (p, α^0) , then $\beta \cdot (\alpha^0 - F(z^*)) = 0$.

- i. Continue to assume whatever it takes to make $\pi(p, \alpha)$ a concave function of α . Suppose we now pose the problem

$$\text{maximize } p \cdot z - \beta F(z),$$

with no restrictions whatsoever on z other than $z \in R^k$, where β is taken from part h. That is, β along with γ provides a supergradient to π in α at (p, α^0) . In words, the firm can do whatever it wants, but it must pay for the resources it consumes (and is paid for resources it “provides”), where β gives the price vector on the n resources. (The term “shadow prices” might ring a bell for you at this point.) Prove that if z^* was a solution to the originally posed problem, it is as well a solution to this “no constraints, but pay for the resources you use” problem.

Assignment #4

There is one problem, numbered 8.

- 8. The answer to many parts of what follows involves citation of basic results from the text, plus (in some cases) verification that certain conditions hold and/or special handling for “boundary” cases. Don’t be surprised when this is true.)

Consider a two-person, two-good, pure-exchange economy. The two persons are Alice and Bob. They have continuous, strictly convex, and strictly increasing preferences $\succeq^A$ and $\succeq^B$, respectively. Their individual endowments will vary in this problem; but the social endowment $e \in R_{++}^2$ never changes. (N.B., the social endowment is strictly positive.)

In the usual fashion, let

$$X := \{(x^A, x^B) \in (R_+^2)^2 : x^A + x^B \leq e\}.$$

Let P be the subset of points in X that are Pareto-efficient allocations of the social endowment, and let

$$W := \{x \in X : x \text{ is the allocation portion of a Walrasian equilibrium } (p, x) \text{ for some prices } p \text{ and some initial allocation } (e^A, e^B) \text{ for which } e^A + e^B = e.\}$$

a. $W = P$. Why?

b. Let u^A and u^B be continuous and strictly concave representations of the preferences of Alice and Bob, respectively. Consider the problem

$$\begin{aligned} &\text{Maximize } F((x^A, x^B), \theta) := \theta u^A(x^A) + (1 - \theta)u^B(x^B) \\ &\text{subject to } (x^A, x^B) \in X, \end{aligned}$$

for the parameter $\theta \in [0, 1]$. Show that this problem has a unique solution for each θ and, if we write $(x^A(\theta), x^B(\theta))$ for the solution at a given θ , that $\theta \rightarrow (x^A(\theta), x^B(\theta))$ is a continuous function.

c. Is it true that

$$W = \bigcup_{\theta \in [0,1]} \{(x^A(\theta), x^B(\theta))\}?$$

Give a proof or a counterexample.

d. Continuing part b, is it true that $\theta \rightarrow u^A(x^A(\theta))$ is an increasing function of θ ? Either prove this is true or give a counterexample.

e. Now consider the economy $\mathcal{E}(\theta)$ for $\theta \in [0, 1]$ where the initial endowment is $e^A = \theta e$ and $e^B = (1 - \theta)e$. For each value of θ , does there exist at least one Walrasian equilibrium for $\mathcal{E}(\theta)$? Give a proof or a counterexample.

f. Let $W(\theta)$ be the set of Walrasian-equilibrium allocations for the economy $\mathcal{E}(\theta)$. Is it true that, for each θ , $W(\theta)$ is singleton? Either prove that it is (in general, with

no more assumptions than have been made so far), or provide a counterexample. A convincing picture will suffice for a counterexample.

g. Show that $\theta \Rightarrow \mathbf{W}(\theta)$ is upper semi-continuous.

h. Is it true that, if $(x^A(\theta), x^B(\theta)) \in \mathbf{W}(\theta)$ and $(x^A(\theta'), x^B(\theta')) \in \mathbf{W}(\theta')$, where $\theta > \theta'$, then $u^A(x^A(\theta)) \geq u^A(x^A(\theta'))$? (That is, does a larger initial share of the social endowment mean more utility in the end, for Walrasian equilibrium outcomes?) Provide a proof or counterexample, and if you decide this is not true, a convincing picture will suffice for a counterexample.